CSA1701-ARTIFICIAL INTELLIGENCE LAB EXPERIMENTS

Program 16 : Python Program to Implement a Neural Network (MLP Classifier)

Aim

To develop a Python program that trains a neural network (Multi-Layer Perceptron) to learn the OR gate and predict its output.

Algorithm

- Import MLPClassifier from sklearn.neural_network.
- Define the training inputs x (all combinations of 0 and 1) and the OR gate outputs y.
- Create the MLP model with one hidden layer of 4 neurons and train it using fit().
- Read two input values (0 or 1) from the user.
- Predict the output for the given inputs and display it.

Program

```
from sklearn.neural_network import MLPClassifier

x = [[0, 0], [0, 1], [1, 0], [1, 1]]
y = [0, 1, 1, 1]

model = MLPClassifier(
    hidden_layer_sizes=(4,),
    max_iter=5000,
    random_state=1
)

model.fit(x, y)

a = int(input("Enter first value (0 or 1): "))
b = int(input("Enter second value (0 or 1): "))

print("Output:", model.predict([[a, b]])[0])
```

Output

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
============== RESTART: C:/Users/shaik/OneDrive/Documents/16.py ==============
Enter first value (0 or 1): 1
Enter second value (0 or 1): 0
Output: 1
>>>
```

RESULT

Thus, the Neural Network (MLP) was implemented successfully using Python.